Three Paradigms of Artificial Intelligence and Their Moral Imaginaries

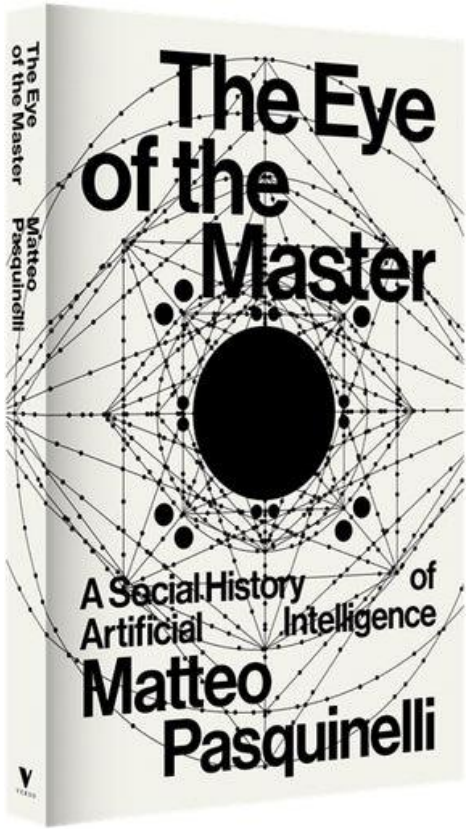
Fernando Pasquini Santos

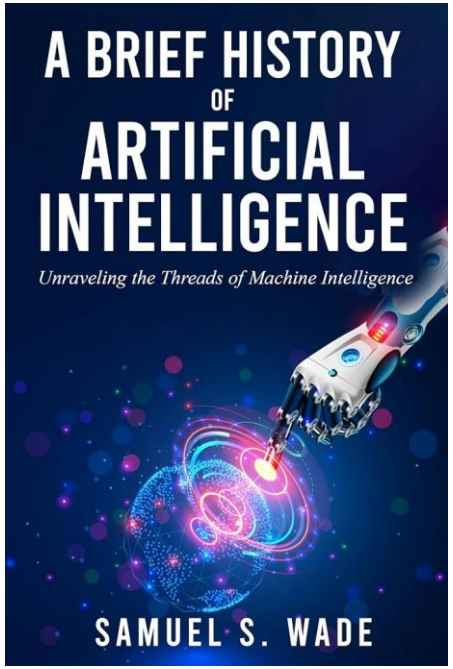
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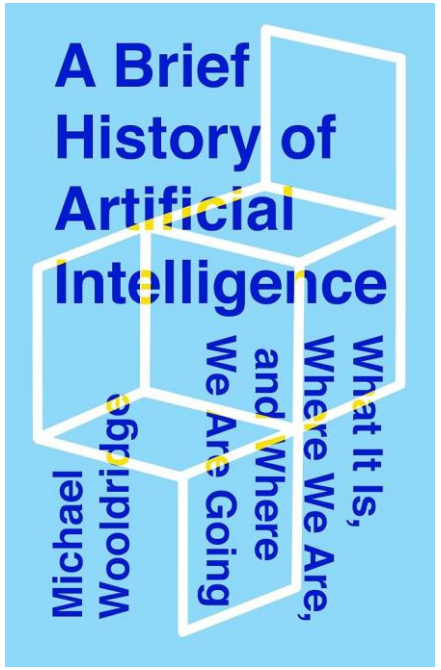
Calvin UNIVERSITY 1876

Navigational Scholarship in AI

- How did we came here? Where are we going? Where to start?
- And how could we integrate all of that in a CS curriculum?







An idea...

AI Phase	Logic	History of Science Parallel
Symbolic (1950-1995)	Deduction	Premodern science (theory)
Adaptive (1995-2020)	Induction	Modern science (experiment)
Generative (2020-now)	Abduction	Computational science (simulation)

Each phase arose from the limitations of the previous…

- **Symbolic AI** revealed the limits of explicit knowledge elucidation (Hubert Dreyfus' critique) → led to data-driven learning.
- **Adaptive AI** revealed the limits of working only with recorded, past data (STS critique) → led to generative, abductive world-exploration.
- **Generative AI** now probably reveals the limits of brute-force optimization and scaling…

Moral Imaginaries and AI

- Decentering the human in “intelligent” operations like **deduction, induction and abduction** leads us to certain moral motifs: delegation, prediction and exploration. All of them can be good, but what are their moral limits?

AI Phase	Logic	**Requires an ethic of…**	**Moral Imaginaries / Motifs**
Symbolic (1950-1995)	Deduction	Delegation	Slavery, The Golem, The Iron Cage
Adaptive (1995-2020)	Induction	Prediction	Divination, Tragedy, The Panopticon
Generative (2020-now)	Abduction	Exploration	Acceleration, Psychedelics, Fragmentation

Ethics of delegation: 1. Slavery

- **“Alienation from life”:** delegation collapses agency and distances us from embodied engagement.
- **Marx / Illich: appropriation & power** — heteronomy through technical systems.
- **Luther’s incurvatus in se** — self-curved inward, enslaved to appetite.
- **Rousseau’s Savage / Stiegler’s “Fault of Epimetheus”** — externalization as loss of essence.



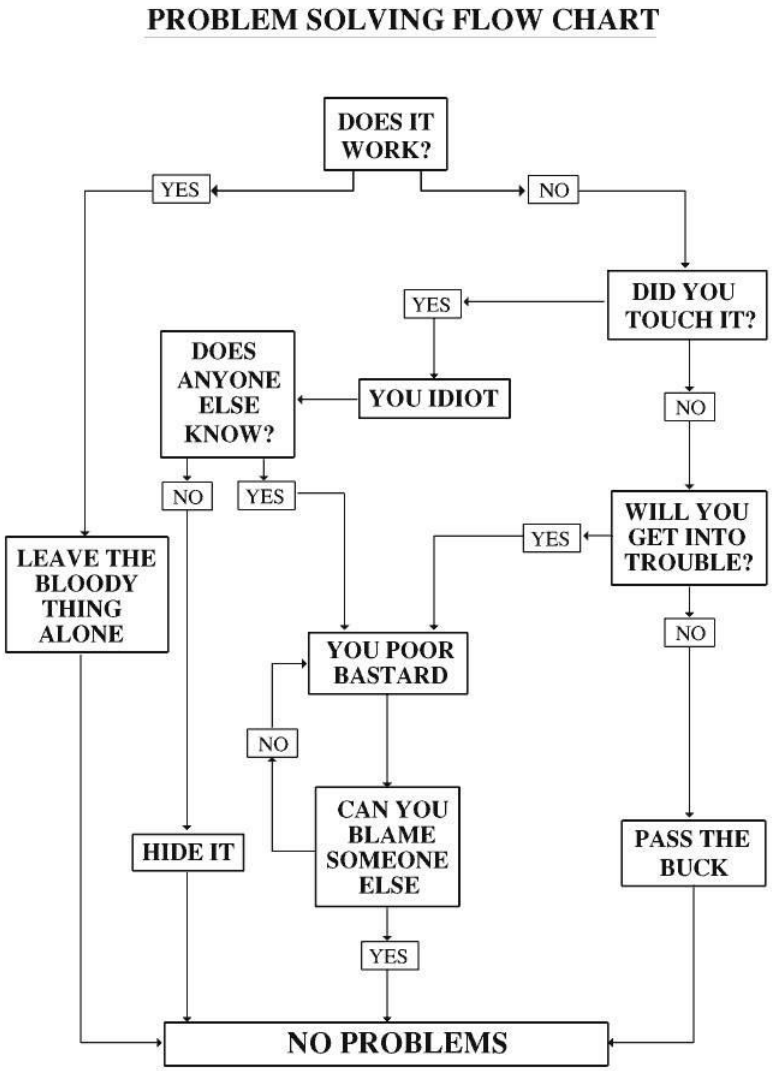
Ethics of delegation: 2. The Golem

- **The Golem from Prague** — Mechanized obedience scaling beyond intention.
- **Talos** — blind automatic execution.
- **The Sorcerer's Apprentice** — outsourcing maturity → chaos.
- **Frankenstein's Creature** — creation surpassing intention without social embedding.



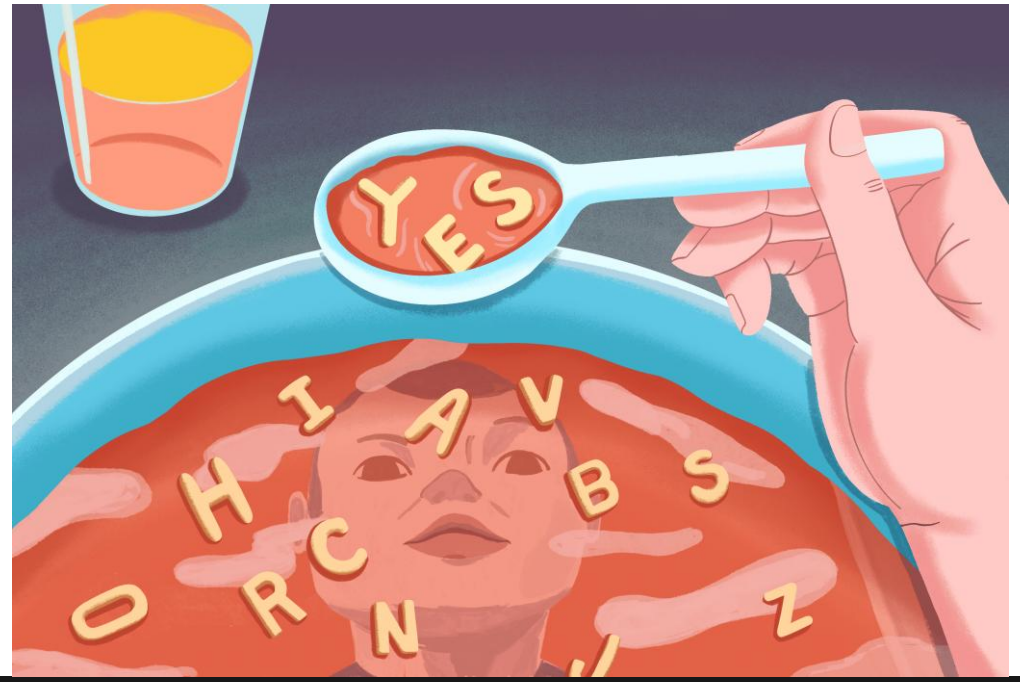
Ethics of delegation: 3. The Iron Cage

- The temptation to reduce wisdom to explicit procedures.
- **Weber's Iron Cage / Kafka** — bureaucratic rationalization displaces meaning.
- **"Legalism** "— mistaking rule-keeping for righteousness; law replaces mercy.
- **Plato's Republic** — totalizing rational schemes for social order.



Ethics of prediction: 1. Divination

- **Augury / Omens** — should we?
- **False Prophets** — saying what people want, not what is true.
- **Cassandra** — anomalies dismissed because they challenge precedent.
- **Apophenia** — seeing patterns where there are none.



Forbes

INNOVATION > ENTERPRISE & CLOUD

Why We're All Practitioners Of Data Divination

By Dan Woods, Former Contributor.

Published Sep 18, 2017, 05:29am EDT, Updated Sep 18, 2017, 05:29am EDT

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We are all practitioners of data divination

JENNIFER WALKER

It's easy for us to look around our world populated by iPhones, AI-driven data analysis,

LOADING VIDEO PLAYER...

Ethics of prediction: 2. Tragedy

- **Past Patterns become Destiny** — what was becomes what must be, future colonized by statistical precedent.
- **Oedipus** — attempts to escape reinforce the pattern.
- **Self-fulfilling prophecies, Pygmalion effect**
- **Great Chain of Being** — hierarchy fossilized as “natural.”



Ethics of prediction: 3. The Panopticon

- Everything must be seen, tracked, and made legible (James C. Scott).
- **Laplace's Demon** – control through total visibility.
- **Eye of Horus** — cosmic order through visibility.
- **Big Brother** — datafied society as total knowledge.



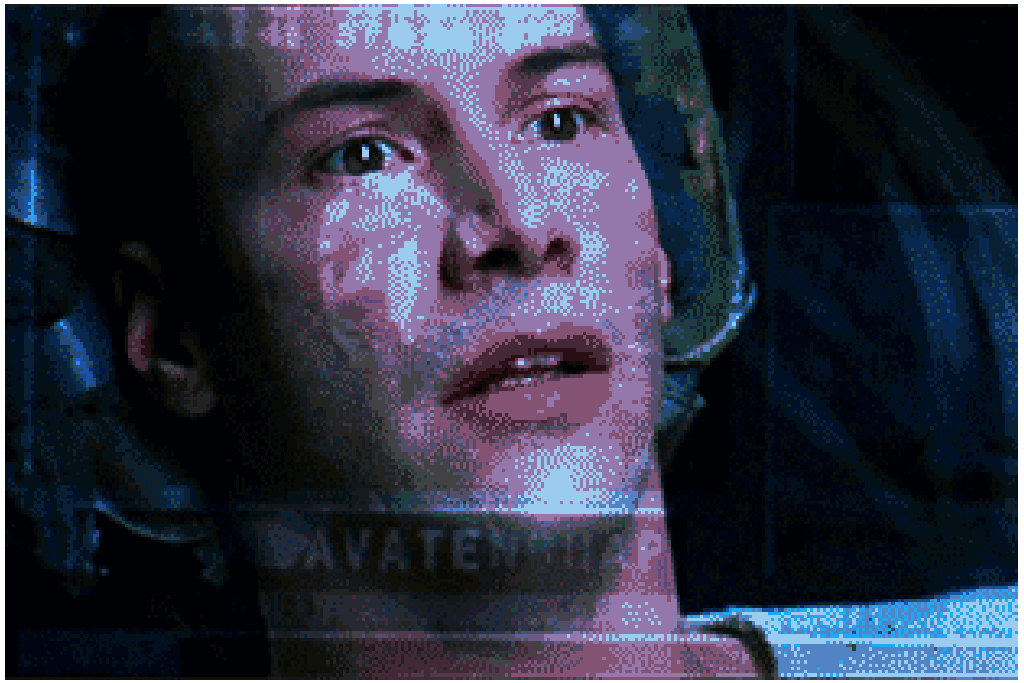
Ethics of exploration: 1. Acceleration

- **Brute-Force Search and Combinatorial Overflow** — Collective creation expands into immense possibility spaces.
- **Gilgamesh's Quest** — exhaustive searching for what cannot be grasped.
- **Scheherazade's 1001 Nights** — generative iteration that proliferates without end.



Ethics of exploration: 2. Psychedelics

- Perception untethered from reality, simulation as experience.
- **Plato's Cave** – mistaking representations for reality.
- **Zhuangzi's Butterfly Dream** – unstable identity and simulation.
- **Galadriel's Mirror** – seeing possible worlds.



Ethics of exploration: 3. Fragmentation

- Local sense, global chaos; attention dilution.
- Possibility explodes outward while interiority thins. “Anything goes”, “nothing really matters”
- **Chimera** — confusing categories and boundary crossing (by mere possibility).
- **Total gamification** — life reduced to reward functions and prompts.



I know, I know

- This is not exhaustive
- Analytically imperfect – categories intermingle, maybe there are more, some other nuances…
- Yet, maybe it is a start.

Virtues and counter-practices

- **Love addresses delegation** with care and engagement - returning the person to the body, the world, and the community.
  - Exodus motif, Drums of freedom (African diaspora), Hartmut Rosa's resonance, Albert Borgmann focal things and practices
- **Faith addresses prediction** by restoring surprise, interruption, promise, and newness.
  - Annunciation/Nativity (Arendt), Antifragility (Taleb), Paul Ricoeur's surplus of meaning.
- **Hope addresses exploration** by restoring proper scale, gift and rest
  - Sabbath, manna in the wilderness, Babette's Feast, Josef Pieper's leisure, Heidegger's Gelassenheit, Ellul's non-power

Some experiences in teaching

- https://inteligenciaartificial.cristaosnaciencia.org.br



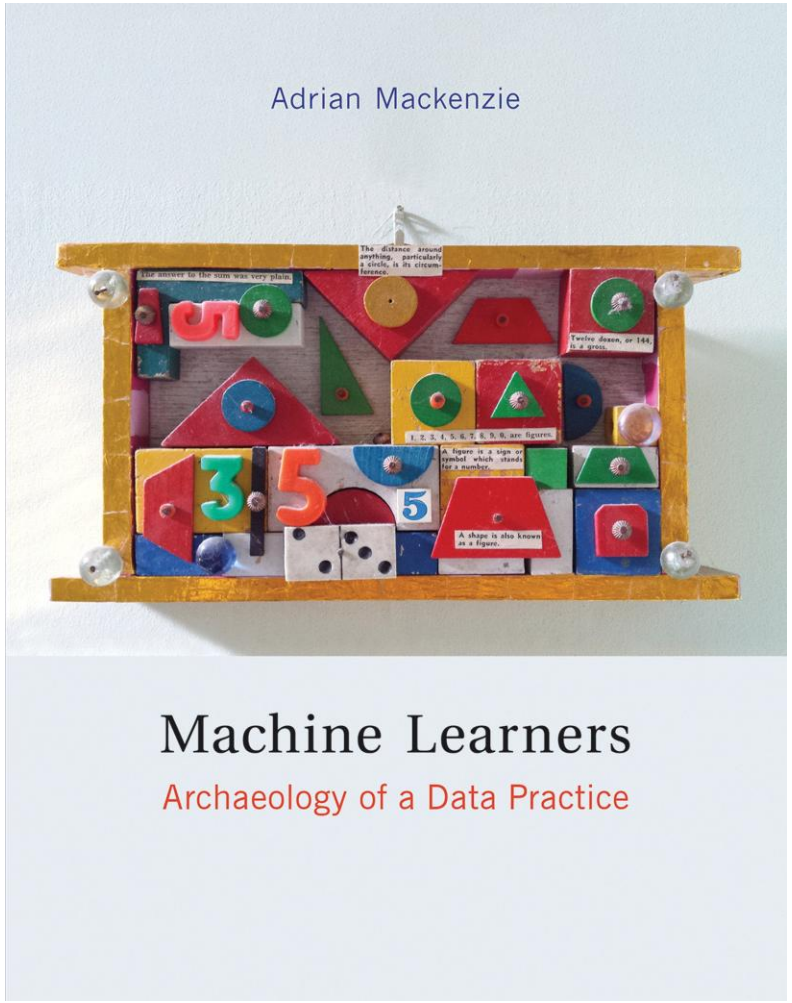
Using media

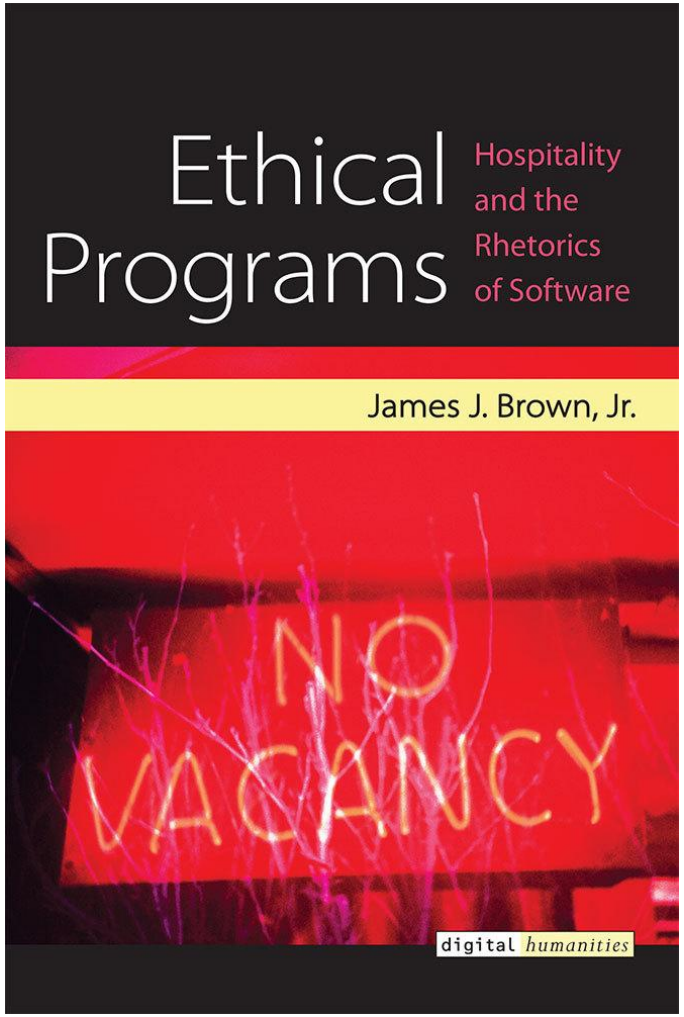
Ethic of…	Motif	Films, series, fiction, games
Delegation	**Slavery**	WALL-E (2008); E. M. Forster, “The Machine Stops” (short story, 1909)
	The Golem	Fantasia, “The Sorcerer’s Apprentice” (1940); Shelley, Frankenstein (1818); Universal Paperclips (browser game, 2017); Black Mirror “Thronglets” (mobile game)
	The Iron Cage	Severance (TV, 2022–); Brazil (Gilliam, 1985); Papers, Please (game, 2013); I, Daniel Blake (2016)
Prediction	**Divination**	Minority Report (2002); Asimov, Foundation (novels / TV); Don’t Look Up (2021)
	Tragedy	Dune (novel/movie); Gattaca (1997); Westworld, season 3 – the Rehoboam system (2020)
	The Panopticon	Eggers, The Circle (novel 2013 / film 2017); Person of Interest (TV, 2011–16); Coded Bias (documentary, 2020)
Exploration	**Acceleration**	Edge of Tomorrow (2014) – learning by endless retries; Borges, “The Library of Babel” (1941)
	Psychedelics	The Matrix (1999); Her (2013); Blade Runner 2049 (2017) – Joi, the simulated companion
	Fragmentation	Everything Everywhere All at Once (2022); Bo Burnham: Inside (2021); Black Mirror, “Nosedive” (2016)

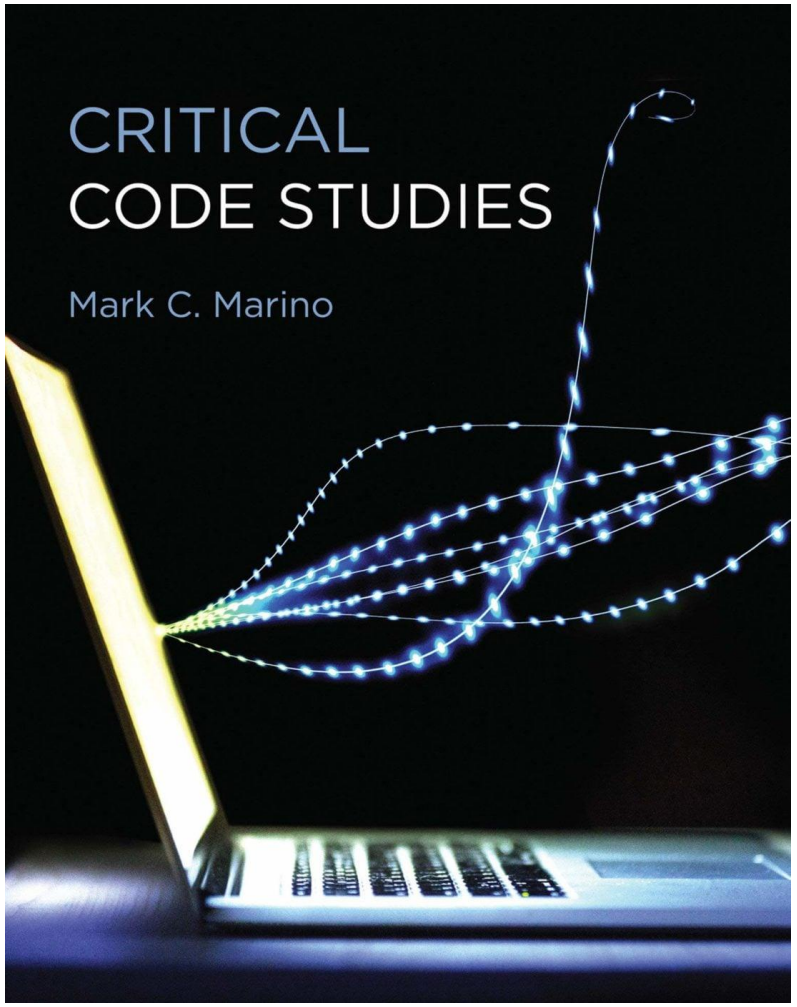
In the technical curriculum

Ethic of…	Motif	Where it can surface in CS courses
Delegation	**Slavery**	Functions & abstraction (“what exactly are we handing off?”); scripting/automation labs; HCI – levels of automation, deskilling
	The Golem	Recursion without a base case, runaway loops; tests vs. intent (the spec is not the wish); reward hacking / specification gaming in ML
	The Iron Cage	If-else constructs as bureaucracies; rule-based expert systems; finite-state machines and form validation
Prediction	**Divination**	Regression & overfitting, train/test split; spurious correlations and p-hacking (apophenia); model calibration
	Tragedy	Clustering algorithms; classifiers trained on historical labels; feedback loops in recommenders and predictive policing
	The Panopticon	Database design (a schema decides what becomes legible); logging & telemetry, cookies and web tracking; privacy & security
Exploration	**Acceleration**	Loops (for/while); reinforcement learning; search & combinatorial explosion (Big-O, backtracking); GPUs & parallelism
	Psychedelics	Computer graphics; simulation & procedural generation; sampling, temperature and “hallucination” in LLMs
	Fragmentation	Embeddings & latent-space interpolation (chimeras); event-driven code & notifications (attention)

Great “integrative” reads







Example

Forum 2: On Coding with Wisdom and Justice

Before class: Read *On Coding with Wisdom and Justice*.

In class: You'll receive a paper with the discussion questions below. In small groups, talk through the questions, annotate the paper with your group's thinking, and compare notes with another group. Before you leave, take a photo of your annotated paper (with everyone's name on it) and upload it to Moodle.

Discussion Questions

1. The article opens with Indiana's welfare system automatically denying over a million aid applications because the algorithm couldn't handle context the way a human caseworker could. Where do you see `if`/`else`-style rules at work in that failure? What context might be hard to capture in a simple condition?
2. Michael Polanyi's idea of *tacit knowledge* is that "we know more than we can tell." Can you think of a decision you make in daily life that would be hard to write as a precise `if` statement, because so much of your judgment is intuitive?
3. Drawing on Wittgenstein, the author argues that "any course of action can be made out to accord with the rule" — meaning rules always need interpretation in context. As you write conditionals this week, where do you find yourself making a judgment call about what a condition "really" means?
4. The author suggests two questions programmers should ask about any algorithm: *what context does this algorithm assume?* and *what conditions would break it?* Pick an app or algorithm you use daily (a spam filter, a recommendation feed, a grade calculator) and try answering both.